

PRACHETA AMARANATH (she/her/hers)

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SUMMARY

My focus is on the intersection of simulation and causal modeling, where I work on constructing interpretable, causal metamodels of discrete-event simulations and using simulation-based techniques to improve causal inference algorithms.

EDUCATION

University of Massachusetts Amherst

Amherst, MA

Ph.D. Candidate

Expected 2026

Dissertation: The Interface of Simulation and Causal Modeling (Advised by David Jensen and Peter Haas)

University of Massachusetts Amherst

Amherst, MA

M.S in Computer Science

2018-2020

Rastreeya Vidyalyaya College of Engineering

Bangalore, India

B.E in Electronics Engineering (Major: Telecommunications)

2011-2015

SKILLS

- **Programming:** Python (expert), R, C++ (familiar), PyTorch, Tensorflow, JAX
- **Machine Learning:** Causal Inference, Deep Learning, Generative Modeling, Probabilistic and Graphical Modeling, Time-series Modeling, Bayesian Inference, Statistics
- **Simulation:** Discrete-event simulation modeling (Simpy, PySim)
- **Systems:** Experience working on HPC clusters (Slurm), Google Cloud, AWS

RESEARCH EXPERIENCE

University of Massachusetts Amherst

Amherst, MA

Graduate Researcher

September 2020 – Present

Science for AI Governance

- Analyzed the use of current methods to identify text written by LLMs in the context of public comments

Army Research Office – Devcom Analysis Center

- Designed a simulation model as a testbed for logistic networks pertaining to the transport of equipment
- Developed a simulation metamodel that can be used for efficient analysis of complex simulation models (P1)

Knowledge Discovery Lab

- Developed a causal metamodel to efficiently analyze conditional and interventional scenarios for Markovian discrete-event simulations (P1)
- Developed a framework using simulation-based inference and generative neural networks for generating synthetic data to evaluate causal effect estimation algorithms (P2)

DREAM Lab

- Developed an algorithm to estimate the prevalence of multiple chronic conditions using maximum entropy, frequent itemset mining and machine learning (P3)

PROFESSIONAL EXPERIENCE

Microsoft Research

Bengaluru, India

Visiting Scholar

November 2024 – January 2025

- Investigated the use of generative methods in evaluating causal effect estimation algorithms

Google X, The Moonshot Factory

Mountain View, CA

Research Intern

June 2023 – January 2024

- Designed and developed a simulation model to empirically evaluate a computational biology pipeline

EBSCO Information Services

Ipswich, MA

Semantic Analysis and Modeling Intern

May 2020 – August 2020

- Developed an algorithm to extract information of population and gender metadata from clinical text
- Improved the article classification capabilities for EBSCO's content database using topic modeling and polysemy detection

Massachusetts Dept. of Public Health
Data Science for the Common Good Fellow

Amherst, MA
May 2019 – August 2019

- Analyzed the impact of social determinants of health to assess risks for communities in Massachusetts

Cisco Systems (India)
Systems Engineer

Bengaluru, India
July 2015 – July 2018

- Responsible for solution design and testing of routing, switching, wireless, data center and network programmability for enterprise customers

PUBLICATIONS

- P1. **Pracheta Amaranath**, Peter J. Haas, David Jensen, and Sam Witty. "[Causal Dynamic Bayesian Networks for Simulation Metamodeling](#)." In 2023 Winter Simulation Conference (WSC), pp. 746-757. IEEE, 2023.
- P2. **Pracheta Amaranath**, Vinitra Muralikrishnan, Amit Sharma, and David Jensen. "[Improving Generative Methods for Causal Evaluation via Simulation-Based Inference](#)." arXiv preprint arXiv:2509.02892 (2025)
- P3. **Pracheta Amaranath**, Ninad Khargonkar, Prasanna Srinivasan, Roshan Thaikkat, Hari Balasubramanian, Peter J. Haas. "Estimating the Prevalence of Multiple Chronic Diseases via Maximum Entropy." (In progress).
- P4. Laura B. Balzer, Erica Cai, Lucas Godoy Garraza and **Pracheta Amaranath**. "[Adaptive Selection of the Optimal Strategy to Improve Precision and Power in Randomized Trials](#)." Biometrics, Volume 80, Issue 1, March (2024)
- P5. Katherine Avery, Jack Kenney, **Pracheta Amaranath**, Erica Cai, and David Jensen. "[Measuring Interventional Robustness in Reinforcement Learning](#)." arXiv preprint arXiv:2209.09058 (2022).

TEACHING EXPERIENCE

University of Massachusetts Amherst
Graduate Teaching Assistant

Amherst, MA

- Introduction to Simulation Fall 2025
- Machine Learning Spring 2023
- Programming with Data Structures Fall 2020

LEADERSHIP AND SERVICE

- Reviewer: TMLR 2025
- Co-Founder., Voices of Data Science., University of Massachusetts Amherst
- Student Steering Committee, Researchers, Educators and Business Leaders of Massachusetts (REBLs), 2019-2020

AWARDS

- Paper on "Causal Dynamic Bayesian Networks for Simulation Metamodeling" was nominated for the **Best Theoretical Paper Award**, Winter Simulation Conference, 2023
- Awarded a "**Dissertation Proposal Writing Fellowship**", Fall 2024, Manning College of Information and Computer Sciences
- Third-rank**, B.E., Rastreeya Vidyalaya College of Engineering, Bengaluru, 2015

CONFERENCE PRESENTATIONS

- (2021) INFORMS Annual Conference: [Estimating the prevalence of Chronic Diseases using the principle of Maximum Entropy](#)
- (2019) Grace Hopper Conference for Women in Computing: Attendee
- (2019) Women in Data Science - Central Massachusetts: Estimating Chronic Diseases using the principle of Maximum Entropy
- (2019) Data Science for the Common Good: Event Showcase (University of Massachusetts, Amherst): Assessing health risks for communities in Massachusetts (in collaboration with Mass. Dept. of Public Health and Data Science for Common Good)